

Data-Driven Strategy for Pricing, Marketing & Seasonal Campaigns at FreshMart

Week 7 — HW5

Group 4

Alina, Ina, Yilin, Yuki

Business Context

*FreshMart is a premium grocery delivery service facing stiff competition. **The company needs to optimize its marketing budget and pricing strategy to maximize sales** while minimizing wasteful spending.*

Data & Methodology

Dataset:

FreshMart_sales_data_sample.csv

Variables included:

- **product_price**: Price of FreshMart's meal kits (\$k)
- **competitor_price**: Competitor's meal kit price (\$k)
- **tv_ad_spend**: Daily TV ad spend (\$k)
- **email_marketing**: Daily email campaign budget (\$k)
- **social_media_spend**: Daily Instagram/Facebook ads (\$k)
- **is_holiday**: Indicator variable (1 = public holiday, 0 = otherwise)

Target Variable: Sales

Part 1: Pricing Strategy

- OLS regression: price, competitor price, ad spend, holiday
- Ridge regression (cv.glmnet, 10-fold CV) to address multicollinearity

Part 2: Marketing Spend Optimization

- Lasso regression (cv.glmnet, 10-fold CV) to identify effective channels
- Stepwise/backward OLS as alternative feature selection
- Compare results: Lasso vs. stepwise

Part 3: Seasonality & TikTok Influencer Impact

- LOESS plots of sales vs. day_of_year (span 0.1–0.6)
- B-splines (manual knots: 100,150,250) + controls
- GAM with s(day_of_year) for automatic smoothing
- Compare models via AIC; plot smooth terms
- Extend best model with viral_campaign + interaction
- Evaluate campaign significance & sales lift timing

Part 1:

Pricing

Strategy

Analysis

Sales Drivers: OLS Regression Model Insights

Predicting Sales with Product Price, Competitor Price, Daily TV Ad Spend, Email Marketing Budget, Social Media Spend, & Indication of Public Holiday

Call:
lm(formula = sales ~ product_price + competitor_price + tv_ad_spend + email_marketing + social_media_spend + is_holiday, data = fresh)

Residuals:
Min 1Q Median 3Q Max
-333.0 -210.9 -121.4 143.6 1352.2

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1172.48499	17.15477	68.347	< 2e-16 ***
product_price	-1.77994	2.27443	-0.783	0.434
competitor_price	0.90570	2.47720	0.366	0.715
tv_ad_spend	0.43783	0.07351	5.956	2.84e-09 ***
email_marketing	0.13917	0.20538	0.678	0.498
social_media_spend	-0.08485	0.16715	-0.508	0.612
is_holiday	193.27494	47.15986	4.098	4.26e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 289 on 3449 degrees of freedom
Multiple R-squared: 0.01659, Adjusted R-squared: 0.01488
F-statistic: 9.697 on 6 and 3449 DF, p-value: 1.298e-10

Observations

- **Price Sensitivity:** +\$1 in FreshMart price → -1.78 units in sales
- **Competitive Advantage:** +\$1 in competitor price → +0.91 units in sales

Implication

Customers are highly price-sensitive;
competitor pricing creates opportunity to capture demand

Strategic Note

Both prices are correlated but critical — FreshMart adjusts tactically to competitor moves, and both are key signals for predictive accuracy in dynamic pricing

Competitor Pricing Impact: OLS vs. Ridge Regression

Uncovering overpricing risks of the OLS model + justifying adoption of stable ridge model

Model	Product Price Coefficient	Competitor Price Coefficient
OLS	-1.780	+0.906
Ridge	-0.682	-0.247

OLS:

Suggests competitor price hikes boost FreshMart sales significantly

Ridge:

Suggests a modest decline in sales instead, a **more conservative, realistic view**

Strategic Takeaways

- OLS model likely inflated due to overlap with FreshMart's own pricing
- Ridge model has more conservative estimate: small decline in sales instead of a large gain

Business Implications

- Over-relying on OLS could lead to overpricing risk and missed demand
- Ridge provides more stable, decision-ready insights for pricing and marketing strategy

Code

Q1:

```
ols_model <- lm(sales ~ product_price + competitor_price +  
               tv_ad_spend + email_marketing +  
               social_media_spend + is_holiday,  
               data = freshmart)
```

```
summary(ols_model)
```

Q2:

```
library(glmnet)  
set.seed(12345)
```

```
preds <- as.matrix(fresh[, c("product_price", "competitor_price",  
                             "tv_ad_spend", "email_marketing",  
                             "social_media_spend", "is_holiday")])  
outcome <- freshmart$sales
```

```
cv_fit <- cv.glmnet(preds, outcome, alpha = 0) # 10-fold CV by default  
best_lambda_ridge <- cv_fit$lambda.min  
print(best_lambda_ridge)
```

Q3:

```
ridge_fit_final <- glmnet(preds, outcome, alpha = 0, lambda =  
best_lambda_ridge)
```

```
ridge_coef <- coef(ridge_fit_final, s = best_lambda_ridge)
```

```
ols_product <- coef(ols_model)["product_price"]  
ols_competitor <- coef(ols_model)["competitor_price"]
```

```
ridge_product <- ridge_coef["product_price", ]  
ridge_competitor <- ridge_coef["competitor_price", ]
```

```
impact_ols <- ols_competitor * 10  
impact_ridge <- ridge_competitor * 10
```

Part 2:

Marketing

Spend

Optimization

Sales Drivers: Lasso Regression Model Insights

Predicting Sales with TV ad spend, social media spend, email marketing spend, control variables, using **Lasso regression (cv.glmnet)** to conclude **which channels should FreshMart continue funding**

Variable	Coefficient	Interpretation
product_price	-0.71	Higher product prices reduce sales (as expected)
competitor_price	0 (excluded)	Not a strong predictor when other variables are included
is_holiday	+165.23	Sales increase significantly on holidays
email_marketing	+0.014	Very small but non-zero effect — contributes marginally
social_media_spend	0 (excluded)	No added predictive value — sales not driven by this
tv_ad_spend	+0.395	Strong, consistent predictor of sales

Continue Funding for:

- TV advertising
- Email marketing during holidays

Cut Funding for:

- Social media spending
- Competitor price tracking

Reallocate Budget to:

more effective TV advertising & email campaigns
+ should intensified during holiday periods

Model Optimization: Iterative Model Selection Insights

Utilizing stepwise & backward selection models to finalize model

Stepwise model

```
Call:
lm(formula = sales ~ product_price + tv_ad_spend + is_holiday,
    data = freshmart)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-334.5	-210.6	-121.3	144.2	1348.8

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1173.98135	15.34294	76.516	< 2e-16	***
product_price	-0.95900	0.42662	-2.248	0.0246	*
tv_ad_spend	0.43773	0.07348	5.957	2.83e-09	***
is_holiday	192.50000	47.12093	4.085	4.50e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 288.9 on 3452 degrees of freedom
Multiple R-squared: 0.01636, Adjusted R-squared: 0.0155
F-statistic: 19.13 on 3 and 3452 DF, p-value: 2.651e-12

Backward model

```
Call:
lm(formula = sales ~ product_price + tv_ad_spend + is_holiday,
    data = freshmart)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-334.5	-210.6	-121.3	144.2	1348.8

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1173.98135	15.34294	76.516	< 2e-16	***
product_price	-0.95900	0.42662	-2.248	0.0246	*
tv_ad_spend	0.43773	0.07348	5.957	2.83e-09	***
is_holiday	192.50000	47.12093	4.085	4.50e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 288.9 on 3452 degrees of freedom
Multiple R-squared: 0.01636, Adjusted R-squared: 0.0155
F-statistic: 19.13 on 3 and 3452 DF, p-value: 2.651e-12

Final Variables Selected

- **Keep:** product_price, tv_ad_spend, is_holiday
- **Exclude:** competitor_price, social_media_spend

Final, Optimized Model

```
lm(formula = sales ~ product_price + tv_ad_spend +
    is_holiday, data = freshmart)
```

Comparing Iterative Model Selection & Lasso Model w/ Best Lambda

Understanding why there are differences in feature selection between the two

Feature	Stepwise & Backward Selected	Lasso Selected	Notes
product_price	✓	✓	Consistent negative predictor of sales
competitor_price	✗	✗	Excluded by both — not driving sales
tv_ad_spend	✓	✓	Consistent positive predictor of sales
email_marketing	✗	✓	Lasso keeps it with small weight; stepwise & backward selection drop
social_media_spend	✗	✗	Excluded by both — not driving sales
is_holiday	✓	✓	Consistent positive predictor of sales

Lasso keeps variable “email_marketing” if it slightly helps with prediction, even if it’s not statistically significant

The two iterative selections drop “email_marketing” as it does not significantly reduce AIC

Lasso's penalty allows it to keep weak predictors if they add value to prediction, while stepwise and backward selection remove them if not statistically significant

Coding

Question 2.1

```
preds2 <- as.matrix(freshmart[, c("tv_ad_spend", "social_media_spend",  
                                "email_marketing", "product_price",  
                                "competitor_price", "is_holiday")])  
outcome2 <- freshmart$sales  
  
set.seed(12345)  
cv_lasso <- cv.glmnet(preds2, outcome2, alpha = 1)  
best_lambda_lasso <- cv_lasso$lambda.min  
print(best_lambda_lasso)
```

Question 2.2

```
final_lasso <- glmnet(preds2, outcome2, alpha = 1, lambda = best_lambda_lasso)  
coefficients_lasso <- coef(final_lasso, s = best_lambda_lasso)  
print(coefficients_lasso)
```

Question 2.3

```
##Stepwise selection  
stepwise_model = step(ols_model, direction = "both")  
summary(stepwise_model)  
  
##Backward selection  
backward_model = step(ols_model, direction =  
"backward")  
summary(backward_model)
```

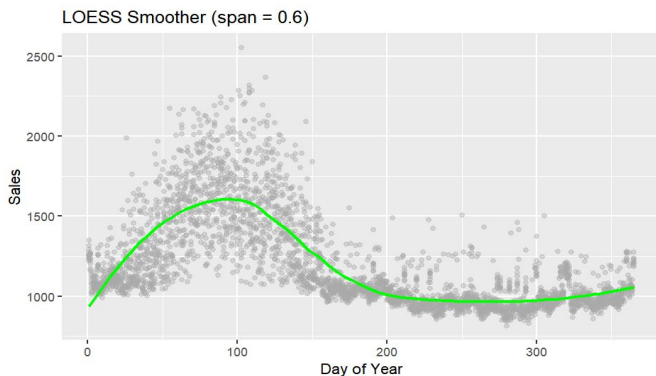
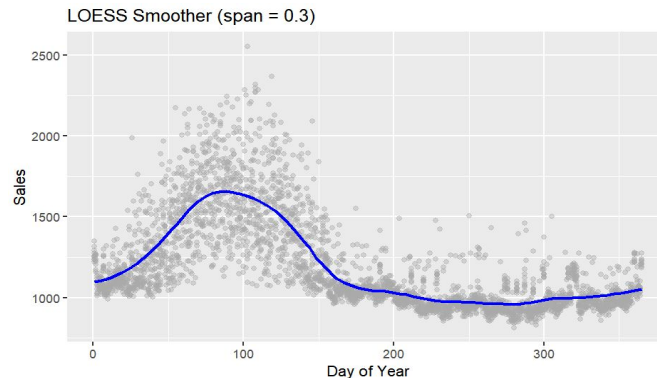
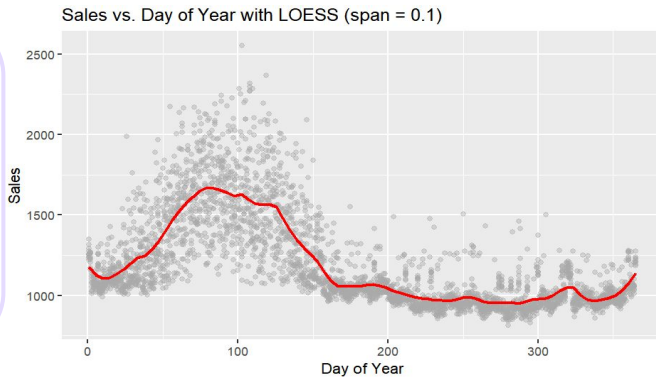
Part 3:

*Seasonality &
TikTok Influencer
Impact*

Exploratory Analysis: Visualizing Seasonal Trends

Creating scatterplot of sales vs. day_of_year with LOESS, spans from 0.1 to 0.6

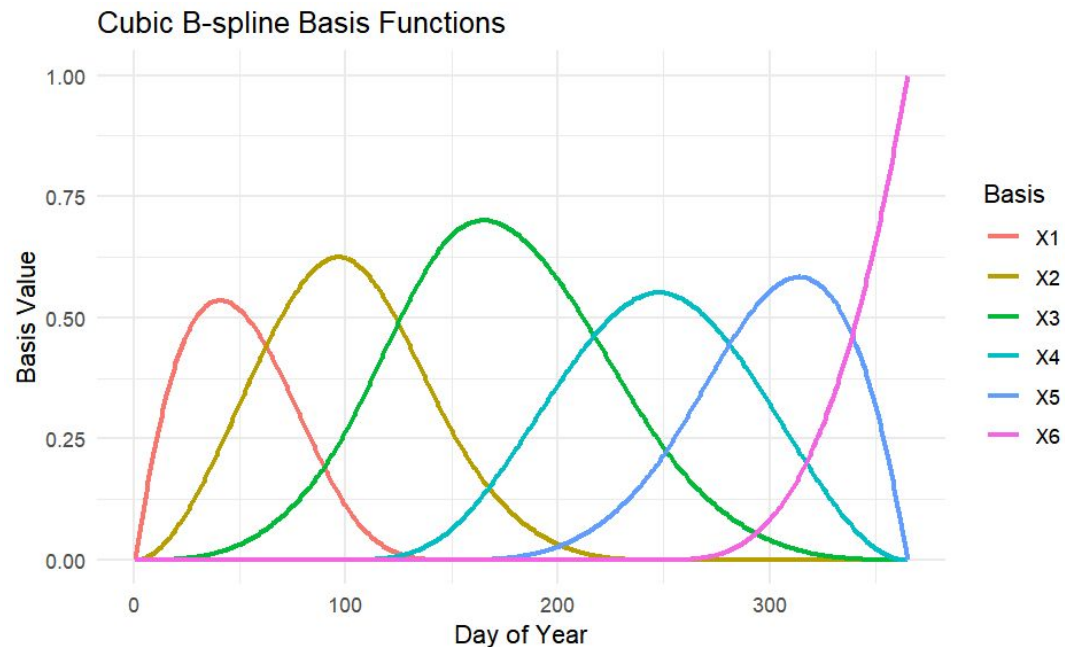
Sales patterns
change around
100th, 150th, and
250th days



In LOESS, a **greater span results in a smoother curve** because it incorporates more data points in each local regression, **making the fit more similar to a global model**

Capturing Seasonal Trends via Generalized Linear Model w/ Manual B-Splines

Capturing nonlinear sales patterns over time, instead of forcing one straight line through all the data



$$\begin{aligned} \text{df} &= k + \text{degree} + \text{intercept} \\ &= 3 + 3 + 0 = 6 \end{aligned}$$

Interpretation: Model will use 6 basis functions to represent the shape of the day_of_year curve

Capturing Seasonal Trends via Generalized Additive Model

Capturing nonlinear sales patterns over time via GAM which automatically chooses the degree of smoothness using cross-validation

Formula:

```
sales ~ s(day_of_year) + temperature + tv_ad_spend + email_marketing +  
is_holiday
```

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	868.21728	8.65554	100.308	<2e-16	***
temperature	17.29924	0.45449	38.063	<2e-16	***
tv_ad_spend	0.41851	0.03402	12.302	<2e-16	***
email_marketing	0.23703	0.09502	2.495	0.0127	*
is_holiday	217.77097	22.21099	9.805	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value	
s(day_of_year)	8.882	8.996	214.4	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.789 Deviance explained = 79%

-REML = 21838 Scale est. = 17865 n = 3456

Interpretation:

Smooth spline with edf ≈ 8.9

The seasonal trend in sales is flexible and includes **multiple turning points** — about **9 degrees of freedom** were used to capture these patterns

Finding Optimal Model to Determine Optimal Campaign Launch Dates

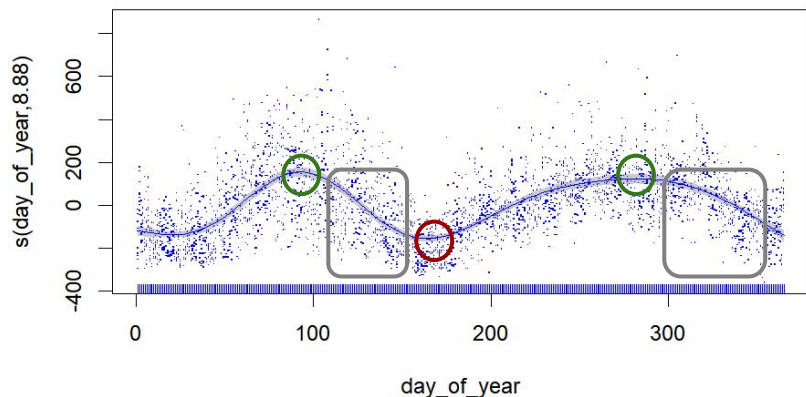
Comparing AICs of B Spline vs. GAM

Model	AIC	Interpretation
B Spline	43687.60	- GAM Model has a lower AIC than the B Spline Model, but the difference is not too different (-27.76) - Conclusion: B Spine Model can be suitable for smaller datasets like this with clear trends, & GAM can be especially useful for more complicated or large datasets with unclear trends.
GAM	43659.84	

Highest predicted sales
on Days 93 & 270

GAM: $s(\text{day_of_year})$

Lowest predicted
sales on Day 164



Slow periods: from ~days 100-150 (post-spring break) & ~days 280-330 (pre-holidays)

Recommendations

Launch campaigns around:

- Days 80-130 → combat mid-year fatigue of sales
- Days 280-310 → re-engage customers before holiday season spending picks up

GLM vs. GAM: Finding Optimal Model to Determine Impact of TikTok Influencer Promotional Campaigns on Sales

$\text{sales} \sim s(\text{day_of_year}, \text{by} = \text{viral_campaign}) + \text{viral_campaign} + \text{temperature} + \text{tv_ad_spend} + \text{email_marketing} + \text{is_holiday}$

Parametric coefficients:

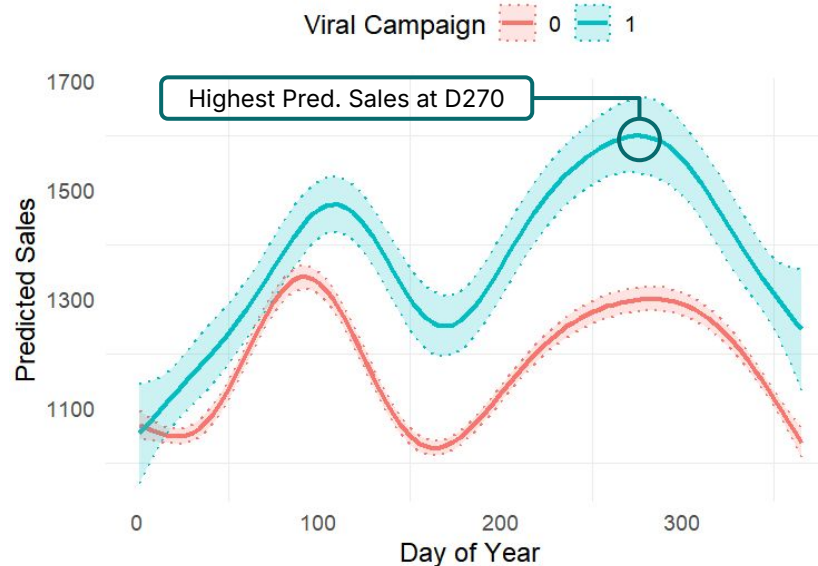
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	860.62880	8.23048	104.566	<2e-16 ***
viral_campaign1	194.89430	10.62302	18.346	<2e-16 ***
temperature	17.18219	0.43149	39.820	<2e-16 ***
tv_ad_spend	0.43508	0.03231	13.467	<2e-16 ***
email_marketing	0.21665	0.09022	2.401	0.0164 *
is_holiday	219.02192	21.09528	10.383	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model Interpretation: The addition of TikTok influencer promotional campaigns (captured by viral_campaign1) **significantly increase projected sales by \$194.89**, $p \sim 0$

Visualization Interpretation: Confidence intervals of the GAM smooth curves for campaign vs. no-campaign show **minimal overlap**, confirming that the **seasonal sales trends are significantly different**

GAM Interaction: Day of Year $\times$ Viral Campaign



Actionable Marketing Insights: Recommendations for Freshmart

Time campaigns with seasonal momentum

- **What to do:** Launch TikTok activations in early spring (Day 80–130) and early fall (Day 250–280)
- **Why:** These windows delivered the strongest sales uplift from influencer promotions

Use influencers to bridge sales valleys

- **What to do:** Target mid-summer (Day 150–220) and late fall (Day 270–320) with influencer partnerships & viral challenges
- **Why:** Helps flatten sales dips and extend customer engagement cycles

Match content to seasonal triggers

- **What to do:** Align TikTok themes with back-to-school, festivals, and spring cleaning
- **Why:** GAM analysis shows seasonal context reshapes sales curves, not just lifts volume

Coding

3.1

```
ggplot(freshmart,aes(x = day_of_year, y = sales)) +  
  geom_point(  
    data = freshmart, # Use original data  
    aes(x = day_of_year, y = sales),  
    alpha = 0.3,      # Semi-transparent  
    color = "black"   # Light gray for background  
  )+  
  geom_smooth(method = "loess", span = 0.3, color = "red", se =  
    FALSE)
```

3.2

```
library(splines)
```

```
knots <- c(100, 150, 250)  
bspline_full <- lm(sales ~ bs(day_of_year, knots = knots, degree = 3)  
+ temperature + tv_ad_spend + email_marketing + is_holiday, data =  
freshmart)
```

```
spline_basis <- bs(freshmart$day_of_year, knots = c(100, 150, 250),  
degree = 3)
```

```
dim(spline_basis)
```

```
bs_basis_df <- data.frame(day_of_year = freshmart$day_of_year,  
bs(freshmart$day_of_year, knots = c(100, 150, 250), degree = 3))
```

```
bs_long <- pivot_longer(bs_basis_df, -day_of_year, names_to =  
"Basis", values_to = "Value")
```

```
ggplot(bs_long, aes(x = day_of_year, y = Value, color = Basis)) +  
geom_line(size = 1.1) + labs(title = "Cubic B-spline Basis  
Functions", x = "Day of Year", y = "Basis Value") + theme_minimal()
```

3.3

```
library(mgcv)
```

```
gam_auto <- gam(sales ~ s(day_of_year) + temperature +  
tv_ad_spend + email_marketing + is_holiday,family = gaussian,  
data = freshmart, method = "REML" # Preferred method for  
smoother estimation)
```

```
summary(gam_auto)  
summary(gam_auto)$s.table
```

3.4

```
AIC(lm_bspline, glm_bspline, gam_auto)
```

```
plot(gam_auto, select = 1, shade = TRUE, seWithMean = TRUE,  
main = "Smooth Effect of Day of Year on Sales", xlab = "Day of  
Year", ylab = "Smooth Function")
```

Coding

3.5 & 3.6

```
library(mgcv)
freshmart$viral_campaign <- as.factor(freshmart$viral_campaign)
gam_interaction <- gam(sales ~ s(day_of_year, by = viral_campaign) + viral_campaign +
  temperature + tv_ad_spend + email_marketing + is_holiday,
  data = freshmart,
  method = "REML")
summary(gam_interaction)
summary(gam_interaction)$s.table

new_data <- expand.grid(
  day_of_year = seq(min(freshmart$day_of_year), max(freshmart$day_of_year), length.out = 100),
  viral_campaign = factor(c(0, 1)))

new_data$temperature <- mean(freshmart$temperature, na.rm = TRUE)
new_data$tv_ad_spend <- mean(freshmart$tv_ad_spend, na.rm = TRUE)
new_data$email_marketing <- mean(freshmart$email_marketing, na.rm = TRUE)
new_data$is_holiday <- mean(freshmart$is_holiday, na.rm = TRUE)

pred <- predict(gam_interaction, newdata = new_data, type = "response", se.fit = TRUE)
new_data$pred <- pred$fit
new_data$se <- pred$se.fit
new_data$upper <- new_data$pred + 1.96 * new_data$se
new_data$lower <- new_data$pred - 1.96 * new_data$se
```

Coding

3.5 & 3.6

```
library(ggplot2)
ggplot(new_data, aes(x = day_of_year, y = pred, color = viral_campaign)) +
  geom_line(size = 1) +
  geom_ribbon(aes(ymin = lower, ymax = upper, fill = viral_campaign), alpha = 0.2, linetype = "dotted") +
  labs(
    title = "GAM Interaction: Day of Year × Viral Campaign",
    x = "Day of Year",
    y = "Predicted Sales",
    color = "Viral Campaign",
    fill = "Viral Campaign"
  ) +
  theme_minimal() +
  theme(panel.grid.major = element_blank(), legend.position = "top", text = element_text(size = 12))

vc_0 <- new_data %>% filter(viral_campaign == 0)
vc_1 <- new_data %>% filter(viral_campaign == 1)

sales_boost <- data.frame(
  day_of_year = vc_0$day_of_year,
  boost = vc_1$pred - vc_0$pred)

max_boost_day <- sales_boost %>% filter(boost == max(boost))
```

Thanks!

Questions or Feedback?